AXIS INTEGRATED RESEARCH PAPERS
Working Paper AXIS2026-M1 (v2) · August 2026 | Pre-Press Methodology Edition

The CC1-4 拼 Methodology

From Component Collision to Clausal Packaging: the Four Pins (四拼) Pipeline

「拼」方法論：從拼件碰撞到拼句包裝——拼音、拼義、拼字、拼枝四拼管線

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Retirement & Consolidation Notice
This paper SUPERSEDES and RETIRES two earlier working papers: AXIS2026-M1 “The Structural Unification of Chinese Morpho-Phonology” (Aug 2026) and AXIS2026-M2 “Component Cascading and the Compression Stack” (Aug 2026). Both remain archived for citation-history purposes but should no longer be cited as current methodology.
What is CARRIED FORWARD unchanged: the CC1-4 four-level architecture, the CCtrio notation (→ Arrow, ‖ Parallel, ❖ Quantum POS Shifter), the Component Density Heuristic, and the 的 (的字) “中華第一字” case study from the old M2.
What is RE-GROUNDED: the old M1's phonological operators (ΨΔR/T, ΨΔC, ΨΔV, framed via Hilbert semantic space and AGI-era language) are replaced by the plainer, citation-first Zero/Tone/Consonant/Vowel-Shift (ZTVC) apparatus set out in §4 below and cross-checked against the SPS Sound Atlas — same underlying claims, less jargon, more testable.
What is NEW here: the Four Pins (四拼: 拼音→拼義→拼字→拼枝) as the master pedagogical pipeline, the M7 clausal assembler (SVOCAm,pp), four full worked case studies (鼓/逗/居/做造製創作), and a fifth self-referential case study applying the whole methodology to its own name (并併拼摒).

Suggested citation: Chan, E.S.L. (August 2026). The CC1-4 拼 Methodology: From Component Collision to Clausal Packaging. AXIS Integrated Research Papers, Working Paper AXIS2026-M1 (v2). DOI: [assigned upon preprint submission].

1. Where M1 Sits in the AXIS Stack

夫子之道，一字以蔽之貫之：拼。（“The Master's Way, summed up and threaded through in a single word: PĪN — to assemble.”）This paper is the methodological engine that sits between two already-published theoretical papers and the applied consumer products (ABC, STORY, SPS, IC) built on top of them. It does not introduce new theoretical claims about Chinese writing; it specifies HOW a learner, or an automated system, actually walks from raw component to finished sentence.

AXIS2026-F1 — WRITING SYSTEMS AS CIVILIZATIONAL TECHNOLOGY

F1 argues that Chinese writing encodes four historical strata simultaneously in every character — an archaeological/semantic layer (MRA, the Macro Radical Alphabet's 26 semantic categories), a phonosemantic bridge layer (300CC, ≈300 Component Clusters), and a modern thematic layer (MRT, Mixed Radical Themes). M1 operationalises this claim: CC1 (Component Collider) is the pedagogical entry point into MRA's semantic categories; the 300CC supercluster families used throughout the ABC/SPS product line ARE the 300CC layer F1 predicts; CC2-CC4's movement from character to clause is the mechanism by which a learner actually crosses from the archaeological layer to the thematic layer F1 describes only architecturally.

AXIS2026-N1 — THE IDEOGRAPHIC CONTINUUM

N1 (“Reconceptualizing Chinese Parallel Compounds through the Character Pair Continuum Framework”) established that many multi-character compounds are not arbitrary pairings but points on a continuum radiating from a shared root — and its Major Update supplement promoted 居 from a minor consonant-shift footnote to a full sub-cluster anchor, with its own T-shift family (踞/倨/鋸/据, jū T1 → jù T4, neutral dwelling escalating to forceful claiming — 擄 is a separate, unrelated character, see Case Study III's correction). M1's CC2 (Parallel ‖ operator) and CC3 (Ideographic Continuum placement) are the teaching mechanism for exactly this kind of finding: N1 discovers and verifies the continuum; M1 packages it into a four-level lesson a learner can actually walk through. Case Study III below (§10, 居) demonstrates this bridge directly.

In short: F1 is the map of the whole territory, N1 is the field survey of specific continuum structures within it, and M1 is the vehicle that gets a learner from one to the other — one component collision at a time.

2. 拼 (pīn) — The Master Verb, and Its Own Best Example

Before defining CC1-4 formally, it is worth noticing that the very word AXIS uses for this methodology — 拼 (pīn, “to assemble, to piece together”) — is itself a textbook case of everything this paper is about: a low-density seed (手 hand + 并 two-people-standing-together) that radiates, through ordinary component mechanics, all the way from “jigsaw puzzle” to “stake your life on it.” A full CC1-4 treatment of this family is given as Case Study V (§13); the short version here motivates why 拼 was chosen as the methodology's name rather than a more neutral term.

手 (hand) + 并 (two people standing side by side) gives 拼 two live semantic potentials from the start: COOPERATE (合併, merge resources) or COMPETE (吞併, annex by force) — the same root component, 并, forks depending on whether the “standing together” is read as alliance or confrontation. A third historical layer adds “staking everything” (from a separate, now-merged character 拚, “to fling the hand open, to give up”), which is why 拼命 means risking one's life, not simply combining objects. One character; three live registers; zero added components beyond the original two. That is the whole thesis of this paper in miniature.

3. The Four Pins (四拼): Pinyin → Pinyi → Pinzi → Pinzhi

CC1-4 (Character Cascades / Collider Combinatorics) is organised around four “pins” — four successive assembly stations a learner passes through, each packaging the output of the one before it. The name 拼 (pīn) itself is used as a bilingual pun: PIN-yin, PIN-yi, PIN-zi, PIN-zhi.

Pin	Name	Function	CC Level
Pin 1	Pinyin 拼音	Acoustic sound & tone register — the ZTVC shift analysis of a character's reading against its family root.	CC1 (phonetic half)
Pin 2	Pinyi 拼義	Visual seed assembly & concept potential — components collide into a character; each collision yields a small set of concrete/kinetic/metaphorical seed meanings (ABC layer).	CC1 (semantic half)
Pin 3	Pinzi 拼字	Lexical colliders & collocation vectors — characters combine into words, phrases, and idioms via the CCtrio operations and the traditional five compound types.	CC2 – CC3
Pin 4	Pinzhi 拼枝（拼枝上幹 + 拼語組句）	Two joint operations: 拼枝上幹 (branches assembled onto the trunk — syntactic tree structure) and 拼語組句 (words wired into full sentences) — carried out together via the M7 assembler.	CC4

拼字 here does not mean “spelling” — Chinese has no letters to spell with. It means character-as-collider: assembling components, then assembling characters, then assembling clauses, through the same underlying combinatorial logic at every scale.

4. CC1 — The Component Collider (拼件 → 拼音 / 拼義)

CC1 has two faces, corresponding to Pin 1 and Pin 2: a PHONETIC face (how a character's reading relates to its family root) and a SEMANTIC face (what concept potentials the visible components generate when they collide).

THE PHONETIC FACE — ZTVC SHIFT ANALYSIS

Superseding the old M1's Ψ-operator formalism, Pin 1 uses the plainer four-way ZTVC test cross-checked against the SPS Sound Atlas: Zero-Shift (identical initial+final+tone — pure semantic labelling), Tone-Shift (tone alone moves — POS/register variation), Consonant-Shift (initial changes via regular alternation — domain jump), Vowel-Shift (final changes substantially — flagged for individual verification, since this is historically the category most likely to hide a false cognate rather than a true one).

Worked Example — the 古 (gǔ) Family under ZTVC
固 (gù, enclosed wall) — Zero-Shift. 錮 (gù, metal lock) — Zero-Shift. Both inherit 古's reading exactly; both are pure semantic-radical additions onto the same sound.
個/箇 (gè, discrete unit) — Vowel-Shift. The final moves from -u to -e; flagged per the standard Vowel-Shift caution, though the “countable, bounded unit” sense plausibly continues 古's “fixed, hardened” core.
涸 (hé, bedrock evaporation) — Consonant-Shift (g→h) with an entering-tone (入聲) stop; a domain jump from “hardened enclosure” to “dried hard,” consistent with the Consonant-Shift = metaphorical-projection pattern used throughout the SPS Sound Atlas.

THE SEMANTIC FACE — PINYI SEED ASSEMBLY

Pin 2 (Pinyi) takes the same character and asks: what concept potentials do its visible components generate when they collide? Each component contributes a “seed potential” — not a single fixed meaning, but a small cluster of live readings (concrete noun, kinetic verb, metaphorical extension) that downstream words will select from. This is the ABC layer already published as the AXIS ABC Card Deck; M1 formalises the assembly rule rather than re-publishing the per-character data.

Formal rule: Pinyi(character) = collide(seed-potentials of each visible component), where the collision output is a RANGE of live readings, not a single gloss — the range is what CC2 then selects from when the character enters a specific word.

5. CC2 — The Lexical Collider (拼字成詞)

CC2 packages CC1's output — characters with their seed-potential ranges — into words. Two independent classification systems apply at this level and are used together: the AXIS CCtrio (carried forward unchanged from the retired M2) and the traditional five-plus-two compound-formation types.

THE CCTRIO — HOW TWO SEEDS COMBINE

Operator	Name	Mechanism
→ / ←	Directional Arrow	One character restricts or generates the other — a head-modifier or verb-object relation (e.g. 打 (strike) + 鼓 (drum) → 打鼓).
‖	Parallel	Two characters sit on the same Ideographic Continuum, radiating from a shared root rather than modifying each other (the mechanism formalised in AXIS2026-N1).
❖	Quantum POS Shifter	A context-forced part-of-speech change with no added component — e.g. 鼓 shifting between noun (the drum), verb (to drum/rouse), and directional intensifier (不斷鼓勵) depending on clausal position alone.

THE TRADITIONAL FIVE-PLUS-TWO COMPOUND TYPES

These sit underneath the CCtrio as a finer-grained classification of the Directional Arrow and Parallel cases specifically, and are used in CC2 teaching material as the standard reference grid:

Type	並列 Coordinate	偏正 Modifier-Head	動賓 Verb-Object	主謂 Subject-Predicate	補充 Verb-Result
Example	語言、開關	核電、微笑	納稅、傷心	地震、年輕	改善、書本

Two further minor types recur across the whole ABC corpus: 附加式 (affixal, e.g. 阿公、房子、作家) and 重疊式 (reduplicative, e.g. 星星、簡簡單單).

6. CC3 — The Collocational Collider (拼配擴詞)

CC3 governs how words expand into phrases and idioms through pre/post-modification, compression/expansion, and figurative collocation — the level at which Chinese starts to diverge most sharply from Germanic modification patterns.

PRE- VS. POST-MODIFICATION

飄逸的長髮 (pre-modification: “flowing hair,” standard descriptive) vs. 長髮飄飄 (post-modification: “hair, flowing,” poetic/classical register, noun stated first, description follows as a reduplicated comment). Both are grammatical; the choice is registerial, not structural — a distinction English marks far more rarely (compare “her flowing hair” vs. the much rarer “her hair, flowing”).

COMPRESSION AND EXPANSION

中央研究院 → 中研院 is the everyday case of CC3 compression; Classical Chinese (古文, informally “估文” — “estimate-text,” text so compressed the reader must reconstruct missing particles) is the extreme case: almost no punctuation, spacing, or function words, forcing the reader to infer part-of-speech and clause boundary from position alone. This compression never fully vanished — it survives as the register speakers reach for when they want maximum density (idioms, classical allusion, formal writing).

FIGURATIVE COLLOCATION — PHYSICAL-TO-ABSTRACT MAPPING

Chinese collocation is unusually figurative: abstract nouns are treated as if they were physical objects, and the verb chosen must match how one would physically handle that object. 抱希望 (hold hope, from 抱著不放, “hug tightly and refuse to let go”) and 提意見 (raise an opinion, from lifting a basket by its handle) are not dead metaphors but active physical mappings a learner can predict once the pattern is named.

7. CC4 — 拼枝上幹 + 拼語組句: Syntactic Branching and Clausal Packaging (the M7 Assembler)

CC4 names two operations that happen together, not one: 拼枝上幹 (assembling branches onto the trunk — the syntactic-tree structure of a clause) and 拼語組句 (wiring words into the finished sentence). AXIS treats these as a single joint pin, Pinzhi, because in practice a Chinese speaker builds both simultaneously — there is no separate step where the “tree shape” is decided before the words are chosen, unlike the more sequential head-first planning typical of Germanic sentence construction. AXIS names the engine that performs both operations M7 — the Magnificent Seven — after its seven possible slot types: Subject, Verb, Object, Complement, Adverbial modifier, plus optional parenthetical phrases (SVOCAm, pp).

The Bamboo vs. the Trunk — a Structural Metaphor
English sentence structure is a TRUNK with BRANCHES: one main clause carries subordinate modifiers and parentheticals hanging off it, usually signalled by connectors (which, because, while).
Chinese sentence structure is BAMBOO WITH SHOOTS: successive clausal segments, often comma-joined with zero connectors, each contributing one M7 slot in sequence. Function words and suffixes appear only when the structure would otherwise be genuinely ambiguous.
This is why 他一掌拍死蟑螂 (“he one-palm slapped-dead the cockroach”) needs no preposition where an English translation needs several, and why classical compression (一鼓作氣 attacking a mountaintop, no preposition for “with”) survives directly into modern usage as 一鼓作氣爬上山頂.

AGENT VS. TOPIC SENTENCES

M7 slots can be organised around an AGENT (a doer performing an action) or a TOPIC (a theme the rest of the sentence comments on) — the same slot inventory, different packaging priority. Case Studies I-II below (§8-9) show both patterns in the same character family.

8. Case Study I — 鼓 (gǔ), Full CC1-4 Walkthrough

Cross-referenced with the SPS Sound Atlas Part II (Drum Roll / Continuous Wave) and the ABC Card Deck. 鼓 is the flagship CC1-4 example because it exercises every pin cleanly.

Pin	Content
Pin 1 · Pinyin	gǔ, tone 214 (dipping) — Zero-Shift anchor for 瞽 (blind musician), Consonant-Shift root for the 彭/澎/膨 Continuous-Wave family (Visual-Semantic Lineage, not phonetic descent — a shared-component-plus-motivated-semantic-extension relationship, distinct in kind from a phonetic-cognate claim; see the SPS Sound Atlas Part II).
Pin 2 · Pinyi	壴 (standing drum on pedestal, seed: hollow/taut/resonant cavity) + 支 (hand gripping a stick, seed: sustained external force / vibration). Collision → concrete noun (the drum), kinetic verb (to strike, drum), metaphorical cascade (rouse, swell, bulge, pluck up courage).
Pin 3 · Pinzi	打 (strike) + 鼓 → 打鼓 (Arrow); 鼓 + 勵 (strive) → 鼓勵 (Arrow, psychological vector); 鼓舞/鼓動/鼓吹 (Parallel, Ideographic Continuum: rousing spirit → inciting a crowd → advocating/hyping).
Pin 4 · Pinzhi	Agent-perspective M7: 「連續不斷的造勢大會，令新上台的領袖感到非常鼓舞，受到鼓勵的他蓄勢待發。」— topic-fronted causative chain, zero connectors between clausal beats, each comma marking one M7 slot handoff.

Note the Quantum POS Shifter (❖) in action: 鼓起 shifts between dynamic verb (鼓起勇氣, actively summoning courage) and stative result (頭上鼓起了一個大包, a bump has swollen up) with no component added — the shift is carried entirely by clausal position and the aspectual particle 了.

9. Case Study II — 逗 (dòu), Full CC1-4 Walkthrough

Cross-referenced with AXIS2026-N1's flagship 戲Ⅱ劇 continuum discussion and the ABC Card Deck. 逗 demonstrates the Parallel (Ⅱ) operator at CC2 particularly cleanly, via a four-way Zidoku-style cross rather than a linear chain.

Pin	Content
Pin 1-2 · Pinyin/Pinyi	辶 (trajectory with crests and troughs — a path that pauses as well as moves) + 豆 (pedestal vessel, seed: object of ceremony/feasting). Collision → a vessel's journey that periodically stops: physically (逗留, to linger), grammatically (逗號, a comma — a written pause), socially (挑逗/逗引, to tease/lure — lingering at the feast becomes flirtation).
Pin 3 · Pinzi (CC2 cross)	Four Arrow-type collocations radiate from 逗 in a cross rather than a chain: 逗 + 引 → 逗引 (luring); 逗 + 弄 → 逗弄 (playful toying); 撩 + 逗 → 撩逗 (sensual hooking); 逗 + 笑 → 逗笑 (eliciting laughter).
Pin 3 · Pinzi (CC3 compression)	百般挑逗不果 (incessant flirting, zero results) — a clamp-compression idiom; 挑三揀四 → coined variant 挑三逗四, substituting the expected 揀 (pick) with 逗 (tease) for comic effect — a live demonstration that CC3 idiom slots are productive, not fixed.
Pin 4 · Pinzhi	Topic-perspective M7: 「貓咪對著專注於工作的主人進行各種撒嬌賣萌，但百般挑逗不果，最後索性直接躺在鍵盤上睡覺。」— Subject / Action / Climax-Conflict / Final-Resolution, four M7 beats in sequence, no subordinating connectors.

10. Case Study III — 居 (jū / jù), the Direct F1↔N1↔M1 Bridge

This is the paper's clearest demonstration that M1 is a pedagogical layer ON TOP OF N1's findings, not a competing theory. AXIS2026-N1's Major Update promoted 居 to a full sub-cluster anchor with its own T-shift family — the case study below is the CC1-4 lesson built directly from that finding, cross-referenced to the fuller EP002 radical-decoding treatment of 尸/古.

Pin	Content
Pin 1 · Pinyin	N1's finding: the 居-family T-shift network runs jū (T1, neutral dwelling) → jù (T4, falling — decisive, forceful claiming): 踞/倨/鋸/据 all carry the assertive T4 reading. 拮据 is flagged in N1 as a cross-family trap — it visually resembles the 吉/古/居 triangle but 吉's phonetic role in 拮 is independent of 居/古's.
Pin 2 · Pinyi	尸 (a person in bent/kneeling posture — semantic) + 古 (phonetic per 說文: 從尸古聲; AXIS additionally notes the 右文說-style homophone bridge to 股/鼓, OPEN and clearly labelled as an AXIS-proposed layer, not a settled reading — see the EP002 radical paper).
Pin 3 · Pinzi	踞 (T-shift, buttocks-on-ground posture) ‖ 倨 (Parallel, the same posture reframed as a character trait — 前倨後恭). Note: 據 (occupy by force) belongs to a different family entirely — the 戲‖劇 tiger-boar continuum (老虎野豬劇鬥以後，漁人得利據死豬為己有) — and is not part of the 居 family. See Case Study VI (§13, 拮据) for 据, the visually distinct character that IS part of this family.
Pin 4 · Pinzhi	STORY-register M7 (matches the EP001 STORY cross-reference tagging convention): 「自古地廣人稀，都是先盤踞，再定居，再固守家園，繁衍後代。」— four sequential M7 beats (occupy → settle → defend → propagate), each one a T4-register verb from the same family.

11. Case Study IV — 做／造／製／創／作, Differentiating Near-Synonyms

The three prior case studies show CC1-4 explaining a single character radiating outward. This case study shows the same apparatus running in the opposite direction: differentiating five near-synonymous “to make” verbs that a rote-memorisation approach cannot distinguish, and predicting which two-character compounds of them are grammatical.

Character	Pinyi Seed Collision	Register / Layer
創	倉 (warehouse/stockpile) + 刂 (knife)	Disruption — cutting a gap out of an existing reserve. First-mover, boundary-breaking.
作	亻 (person) + 乍 (sudden onset)	Epiphany — a spontaneous, top-down burst of inspiration from the individual.
製	制 (cut to standard/protocol) + 衣 (garment/template)	Standardisation — protocol layer, defines repeatable specification for mass output.
造	告 (report/align to standard) + 辶 (movement, workflow)	Pipelining — multi-stage, multi-site assembly and progression toward a result.
做	亻 (person) + 故 (established precedent/routine)	Routine execution — runtime layer, repeating an existing pattern by hand.

Why 製做 Is Not a Word — a CC2 Combinatorics Argument

製 sits at the ARCHITECTURE/PROTOCOL layer (defines specification and tolerance); 做 sits at the RUNTIME/EXECUTION layer (individual manual repetition). Forcing them together jumps directly from “mould specification” to “person's daily fidgeting” with no intervening layer — an abstraction-level mismatch, not a mere collocation gap.

The system instead resolves each real-world need through its correctly-paired neighbour: industrial/systematic production pairs 製 with 造 (製造: protocol + assembly/scaling); everyday inspired or manual output pairs 作 or 做 with concrete objects (制作/做工: inspiration/output + hands-on execution). 製做 is redundant twice over — neither fully architectural nor fully colloquial — and Chinese usage has pruned it accordingly.

12. Case Study V — 拼／拚／并／併／摒, the Self-Referential Proof

§2 introduced this family as motivation; this section gives it the full CC1-4 treatment, because a methodology that cannot explain its own name has a credibility problem.

TWO MERGED HISTORICAL LINEAGES

拼 (pīn) — 手 (hand) + 并 (two people standing side by side): physically arranging separate pieces together (拼合, 拼湊, 拼圖, 拼貼). 拚 (pàn/pīn) — 手 (hand) + 弁 (originally distinct): to turn the palm over, to fling away, to give everything up (拚死, 拚命 — literally "relinquish one's life"). Because 弁 and 并 are graphically similar, and Mandarin sound-change merged their readings, 拚命 was gradually re-written as 拼命 — one character absorbing two historically separate semantic estates.

THE COMPONENT DENSITY CHAIN — FOUR ESCALATING LAYERS

Layer	Mechanism	Example
1 · Physical Assembly	Fragmented physical pieces joined by hand.	拼圖、拼貼、拼湊
2 · Power Concentration	并's "standing together" read as pooled effort/resources.	拼力、拼勁
3 · Collision / Combat	Two forces meeting head-on at a single point.	拼殺、拼刀、拼搏
4 · All-In / Total Stake	The body/life itself thrown in as the last chip — 拚's inherited sense.	拼命

并'S OWN QUANTUM SHIFT — FROM NOUN-ISH ROOT TO ADVERBIAL CONJUNCT

并 additionally shows a pure Quantum POS Shifter (❖) case with no component change at all: as a bound root it means "stand together / merge" (合併, 吞併 — COOPERATE vs COMPETE, the same root forking on context); as a free adverb/conjunct it becomes 並/并且 ("moreover") or 並駕齊驅 ("side by side, at equal pace") — with 竝 (the archaic double-person form) supplying 並's modern shape. 併 specialises further still into a pure classifying radical rather than a stand-in for a grammatical person, matching 屏's own radical-specialisation pattern.

THE LOOP CLOSES: 并 → 併 → 拼 → 摒

拼 and 摒 are not extra meanings bolted onto this family — together they complete the family's own logic, turned back on language itself, in four stages, not three. 并 first COMBINES (merge separate pieces into contact); 併 then DISCIPLINES that combination into a fixed, specialised structural role (no longer raw contact, but a proper part with a defined function); 拼 then ASSEMBLES the disciplined pieces into an actual finished, working whole — this is the family's own master verb, the literal act of building a character, a word, or a sentence out of components that have already been combined and disciplined; only once something is actually assembled can 摒 do its job, DISCARDING whatever the finished assembly reveals to be redundant. Applied to prose style, this gives 摒棄囉嗦歐化語言，潔淨中文 ("discard verbose Europeanised phrasing; keep Chinese clean") — the same CC3 compression principle (§6) and CC4 connector-optional bamboo structure (§7) restated as an editorial instruction: a connector, preposition, or subordinate clause borrowed wholesale from a foreign sentence pattern is not automatically an improvement; once the sentence is actually assembled (拼) and the native structure is shown to already recover the relation without it, 摒 says cut it.

AXIS Corollary — the 并併拼摒 Loop Is Not Chinese-Specific
The same four-stage loop (combine → discipline → assemble → discard-the-redundant) applies to any language absorbing structure from whichever language currently functions as the world's lingua franca. English holds that role today, exactly as Literary Sinitic once held a comparable role across East Asia — but functioning as the world language is a fact about REACH, not a claim about which language's internal grammar is most economical.
The corollary this paper draws from 并→併→拼→摒: borrowing new CONCEPTS from a dominant world language is combination (并) — healthy, ordinary contact. Refining a borrowed concept into a role that actually fits the borrowing language's own grammar is discipline (併). Actually building fluent,

> working sentences with it — not just holding the borrowed pieces separately — is assembly (拼). Only adopting the source language's VERBOSE SYNTACTIC SCAFFOLDING wholesale (redundant connectors, prepositions, and subordinators the borrowing language's own grammar does not require) skips past discipline and assembly straight to bloat, and is exactly what 摒 exists to prune back out. Each language should uphold the compression and clarity native to its own structure while still participating fully in global exchange — 英語為世界語言，各語言仍應各存其美 (“English may be the world's language; every language should still keep what is its own”).

The result: one two-component seed (手+并) legitimately radiates from “jigsaw puzzle” to “stake your life on it” to a full editorial principle for language contact in general — without ever adding a component, and with the paper's own master verb, 拼, sitting as a load-bearing stage inside its own family's logic. This is the single cleanest illustration in the whole AXIS corpus of why the Component Density Heuristic predicts high combinatoric power for low-density seeds.

13. Case Study VI — 拮据 (jié jū), and Why English Says the Same Thing With a Body

This case study does three jobs at once: it corrects the 據／据 confusion flagged in Case Study III (§10), it gives Pin 3's Germanic-bridging ambition (§6) its strongest worked example yet, and it surfaces a new device — the Twin-Idiom Bridge — for cross-linguistic matches that run morpheme-by-morpheme rather than word-by-word.

FIRST: 據 (JÙ) AND 据 (JŪ) ARE NOT THE SAME CHARACTER

Traditional Chinese keeps these visually distinct, though not phonetically unrelated. 據 means to occupy, to seize, to rely on evidence — its own family belongs with 劇 on the 戲‖劇 tiger-boar continuum (老虎野豬劇鬥以後，漁人得利據死豬為己有, "after the tiger and boar exhaust each other fighting, the fisherman profits by seizing the dead pig as his own"). 据 is a visually distinct character restricted to describing a hand seized up or cramped from overwork — its own base reading is the same jù, but by long-standing convention it is read jū specifically and only inside the fixed binome 拮据, nowhere else. Simplified Chinese erases even the visual distinction: 據 simplifies directly to 据, so the two become completely identical in both writing and sound, leaving no cue at all that 拮据's 据 carries a special reading.

Pin	Content
Pin 1-2 · Pinyin/Pinyi	拮 (jié): 扌 (hand) + 吉 — 說文：「拮，手口共有所作也」(hand and mouth both labouring). 据 (jū): 扌 (hand) + 居 — 說文：「据，撠挶也。从手，居聲」, where 撠挶 (jǐ jū) is the classical medical term for limbs stiffened and unable to flex from overexertion. 居's own seed (a person seated, motionless — 尸+几 in its older form 凥) contributes a "locked in place, unable to move" sense on top of its phonetic role — the hand doesn't just tire, it seizes up and stops.
Pin 3 · Pinzi (original sense)	詩經·豳風·鴟鴞：「予手拮据，予所捋荼」— a bird's hands (claws) and mouth both worked raw building its nest. 唐代杜甫：「親賢病拮据」— extended to friends worn down by hardship and illness.
Pin 3 · Pinzi (modern sense)	手頭拮据 (hard up, financially strained) — money so tight that every transaction jams: hands that would otherwise move fluidly (spending, paying, providing) seize up for lack of resource, the same physical image extended from labour to liquidity.
Pin 4 · Pinzhi	《紅樓夢》第 114 回：「但是手頭不濟，諸事拮据」— parallel clauses, not one merged idiom: "no money on hand" causes "everything jams up." Note: 拮抗 (antagonism, a modern loan-coinage from biology/pharmacology) is sometimes read backward as proof that 拮 has no economic sense — but 拮据's economic sense is independently attested centuries earlier, in Du Fu, so the inference runs the causation backwards (倒果為因).

The Cross-Linguistic Bridge — 拮据 and "Live From Hand to Mouth"

Two unrelated language families independently built the same idiom out of the same two body parts. 拮据's founding image is a bird's HAND (claw) and MOUTH both labouring nonstop to build a nest — no rest, no surplus. English's "hand to mouth" is a resource moving directly from HAND to MOUTH the instant it's acquired, with no pause for accumulation. Both idioms encode the identical underlying claim: without a gap between acquisition and consumption, no capital can ever form, and the resulting state is precarious by construction — not by bad luck.

A SECOND, SHARPER LAYER: THE MATCH RUNS MORPHEME-BY-MORPHEME, NOT JUST WORD-BY-WORD

The bridge goes deeper than one compound matching one idiom. 拮据's two morphemes each independently match a SEPARATE English financial expression: 拮 (the hand-and-mouth labour cycle, no accumulation ever forms) maps onto "hand to mouth"; 据 (a hand seized up, frozen, unable to move) maps onto "crunch" — as in a CREDIT CRUNCH, the point where the flow of money physically seizes to a halt. Read this way, 拮据 is not one image but two, wired together in PROCESS → RESULT order: 拮 describes the ongoing precarious cycle, 据 describes what happens when that cycle finally locks up.

English needs two separate idioms to say what Chinese compresses into one two-syllable word.

Formal Definition — the Twin-Idiom Bridge (拆件雙橋)
A Twin-Idiom Bridge holds when a multi-morpheme Chinese compound's individual components EACH independently match a separate, complete idiomatic expression in another language, rather than the whole compound matching one single foreign phrase. Discovering one is itself informative: it reveals that the Chinese compound has its own internal structure (here, Process + Result) that a single whole-word gloss (“financially strained”) would otherwise flatten and hide.
This is a new, more granular instrument than Pin 3's general Germanic-bridging principle (§6) — that principle matches a Chinese unit against a foreign unit; a Twin-Idiom Bridge specifically requires the foreign side to ALSO need two separate expressions to cover what Chinese fits into one word, making it direct evidence of the Classical Compression principle (§6, §14) operating across languages, not only within Chinese.
Status: a genuine AXIS-original cross-linguistic observation, offered as a teaching parallel — not a claim of historical contact or borrowing between 拮据 and either English idiom. Reusable wherever a future 300CC entry shows the same two-matches-two structure.

14. Why AXIS Insists on Traditional Chinese — the 拮据／據 Case as Exhibit A

拮据 (jié jū, financially strained) uses 据 — a Traditional character whose sole documented role, per 說文解字 (「据，戟挶也。从手，居聲」), is describing a hand seized up from overwork. 據 (jù, to occupy, to rely on — 根據, 佔據, 據為己有) belongs to a completely different family (the 戲‖劇 tiger-boar continuum, §10). In Traditional Chinese the two are visually distinct, though not phonetically unrelated: 据's own base reading is the same jù, and it is read jū only by long-standing convention, only inside the fixed binome 拮据.

Simplified Chinese removes even that visual distinction: 據 simplifies directly to 据, so the rare, family-specific character and the common, everyday character (根据, 依据, 占据) become completely identical in both writing and sound. The one remaining cue that could tell a careful reader "this might be a special case" disappears entirely.

⚠ RED ALERT — Internal Correction, Not a New Finding

This paper's own first draft made exactly this error, twice — once listing 據 as a 居-family member in §10, once misquoting Du Fu with 據 instead of 据 in §13. Both were caught and corrected on review, not by the drafting process itself.

The reason is instructive rather than embarrassing: Simplified-script text vastly outweighs Traditional-script text in most training data, so a rare, family-specific reading gets statistically swamped by the far more common one. A drafting tool defaulting to the wrong reading, and needing a human check against a Traditional dictionary to catch it, is a direct demonstration of the cost of building analysis on merged orthography — for machines and for people equally.

THIS IS NOT A NEW PROBLEM — SIMPLIFICATION IS A SECOND WAVE OF AN OLD ONE

AXIS has already documented one historical instance of this exact failure mode: Clerical Flattening (隸變合流), the Qin/Han-era standardisation that visually merged distinct ancestral graphs — 士/王/壬 collapsing toward a shared modern shape, 豊(lǐ) and 豐(fēng) becoming indistinguishable in print despite descending from unrelated ritual-vessel and abundant-harvest pictographs. Twentieth-century Simplification is not a different kind of event — it is the same character-conflation mechanism, applied deliberately and at far greater scale, by policy rather than by centuries of gradual scribal drift. 據/据 is simply the newest instance of an old, well-attested category of error.

WHY THIS IS A METHODOLOGICAL REQUIREMENT, NOT A PREFERENCE

CC1's entire claim (§4) is that visible components carry real, recoverable semantic and phonetic information — Pinyi's seed-potential collision only works if the components on the page are the actual historical components, not a modern policy-merged substitute. Every AXIS finding that depends on distinguishing two similar-looking characters (豊/豐, 士/王/壬, and now 據/据) is a finding that Simplified orthography actively erases at the source. This is why AXIS's base working script is Traditional Chinese throughout: not nostalgia, not politics, but the plain requirement that the method needs the components it claims to be analysing to still be visibly distinct.

15. Retained from the Retired M2 — the 中華第一字 (的) Paradox

The retired AXIS2026-M2 developed a full case study of 的 (de) as “China's #1 character” — simultaneously the most frequently used AND most frequently omitted character in the language, and historically a concrete noun meaning “target.” This finding is fully retained and now sits inside CC3-CC4 as the canonical example of the Inflection Visualizer: Chinese substitutes external, optional pre/post-inflectors (的, 了, 過, 著) for the internal inflectional morphology (-ed, -ing, -s, -fy) Germanic languages carry inside the word itself. Full technical detail remains in the M2 archive; this paper cites it rather than reproducing it, to avoid duplicating already-validated content under a new label.

16. Status & Flags

Following the evidentiary-labelling convention used throughout the AXIS series (the SPS Sound Atlas and the EP002 radical paper): every claim in this paper is tagged by how settled it is.

Item	Status
CC1-4 four-level architecture	CONFIRMED — carried forward unchanged from AXIS2026-M2, now the organising spine of this paper.
CCtrio (→ ‖ ❖) and Component Density Heuristic	CONFIRMED — carried forward from AXIS2026-M2; empirical corpus-frequency validation remains a stated future test, not yet run.
ZTVC Shift Analysis (Zero/Tone/Consonant/Vowel)	CONFIRMED — apparatus set out in §4 of this paper and cross-checked against the SPS Sound Atlas, replacing the retired M1's Ψ-operator notation.
Four Pins (拼音/拼義/拼字/拼枝) pipeline	NEW — first formalised in this paper; not yet published elsewhere.
M7 Clausal Assembler (SVOCAm,pp)	NEW — first formalised in this paper; worked examples given, formal corpus testing not yet conducted.
居's 古=右文說 homophone-bridge reading (Case Study III)	OPEN — AXIS-proposed, pending Old Chinese reconstruction check; presented with AXIS's own reading attached per the standing disclosure rule (see the SPS Sound Atlas Part II and the EP002 radical paper).
拼／拚 merger history (Case Study V)	CONFIRMED — standard, citable philological account of the 拚→拼 graphic/phonetic convergence.
拮据／據 confusion and the Traditional-Chinese methodological requirement (§13-14)	CONFIRMED — 說文解字-attested distinct characters; demonstrated live by two drafting errors in this paper's own first pass, corrected on review.
Twin-Idiom Bridge (拮=hand-to-mouth, 据=credit crunch; §13)	NEW — first formalised in this paper; offered as a teaching parallel, not a historical-contact claim.

17. Conclusion — 拼 as a Compression / Decompression Engine

Classical Chinese is, among other things, an extraordinarily compressed encoding: a handful of low-density components, combined through a small, reusable set of operations, carrying meaning across four orders of magnitude — from pictorial stroke to civilisational compound — with almost no explicit grammatical scaffolding. AXIS's contribution is not to invent this compression but to give it a name and a decoding pipeline: F1 maps the territory this compression operates across; N1 verifies specific continuum structures inside it; M1 is the four-pin engine that takes a modern learner from raw component to fluent sentence by running that same compression forward, and — when needed for teaching — backward. 拼 is both the object of study and, fittingly, the name of the method.

CLASSICAL COMPRESSION, MODERN CONCISENESS — THE AXIS DEFINITION OF GOOD CHINESE

CC3's compression principle (§6) and CC4's connector-optional bamboo structure (§7) converge on a single working definition: GOOD CHINESE IS CHINESE THAT OMITS A SUFFIX, PREPOSITION, OR CONJUNCTION WHENEVER THE CLAUSAL POSITION ALREADY MAKES THE RELATION RECOVERABLE — adding one only when its absence would create genuine ambiguity. Classical Chinese is the limit case of this principle (maximum compression, minimum scaffolding); modern conciseness is the same principle applied with a lighter touch, and the two registers are read side by side, not as opposites.

Two living modern genres make this concretely visible: 金庸小說 (Jin Yong's novels), whose prose is often praised precisely for smuggling classical density and rhythm into fully modern, colloquially readable fiction; and 公函尺讀 (formal official correspondence), whose registerial economy strips out exactly the connectors CC4 identifies as optional. Both are functioning proof that the classical register is not a museum piece — it is a live resource fluent writers still draw on. This is also the practical answer to a question every AXIS learner eventually asks: why do Hong Kong, Taiwan, and Mainland China all keep 文言文 (Classical Chinese) on the curriculum, despite near-universal student resistance? Because the compression patterns it teaches are not obsolete — they are the same patterns that make 金庸's fiction and formal official prose read as GOOD modern Chinese rather than merely correct modern Chinese. CC1-4 gives this intuition a name and a mechanism, rather than leaving it as an unexplained stylistic preference.

Case Study V's 并→併→拼→摒 loop (§12) generalises this beyond Chinese: English's current role as the world's lingua franca is a fact about reach, not a verdict on which language's grammar is most economical. AXIS's closing methodological stance is therefore symmetrical in four stages — combine freely across languages (并), discipline what is borrowed into a structural role that actually fits (併), actually assemble it into fluent working use rather than leaving it as inert borrowed material (拼), and only then discard whatever verbose scaffolding turns out not to be needed (摒). Chinese should stay clean of unnecessary 歐化 (Europeanised) scaffolding for the same reason English speakers benefit from noticing their own default verbosity: every language keeps what is its own, even while speaking to the world.

Author Identifiers & Distribution Channels

All AXIS research, datasets, and consumer products are published under a single consistent identity — Eddie SL Chan — across the following channels, separated by function.

ACADEMIC IDENTITY & PREPRINTS

Channel	Link
ORCID	https://orcid.org/0009-0009-8990-9504
SSRN — Ideographic Continuum (N1)	https://papers.ssrn.com/sol3/papers.cfm?abstract_id=7153478
SSRN — 300CC (F1 dataset)	https://papers.ssrn.com/sol3/papers.cfm?abstract_id=7147918
LingBuzz — 300CC	https://ling.auf.net/lingbuzz/010162
LingBuzz — Ideographic Continuum	https://ling.auf.net/lingbuzz/010165

DATASET & PRODUCT PORTALS

Gumroad and Patreon host the applied consumer products (ABC Card Decks, STORY packs, SPS Sound Atlas, and patron-tier Ideographic Continuum datasets) built on top of the methodology formalised in this paper — the empirical layer feeding back into future AXIS papers.

SOCIAL & COMMUNITY CHANNELS — THE LIGHTER SIDE OF THE RESEARCH

Research findings are also shared in a more informal, user-facing register — written from the perspective of the actual learners using these characters day to day — across Substack, LinkedIn, X, Reddit, PTT (Taiwan), and chinese.stackexchange.com, all under the same Eddie SL Chan identity to keep the academic and public-facing threads traceable to one source.

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